

Regime-Aware Sector Rotation with Momentum and Canary Assets

Research summary of the model, methodology, and conclusions

Arthus Goujon

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This note summarizes a Python empirical research project that reproduces and extends a regime-aware Hybrid Asset Allocation (HAA) market-capitalization rotation paper. The extension tests whether the same HAA architecture can be applied to U.S. sector ETFs and whether the resulting strategy remains credible after walk-forward selection, bootstrap inference, canary robustness checks, and ETF proxy factor regressions.

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1 Executive Summary

The project studies a rule-based tactical allocation model. It starts from Richman’s HAA-style market-capitalization rotation framework and extends it from large/mid/small-cap ETF rotation to U.S. sector ETF rotation. The reference framework uses a multi-horizon momentum signal, a canary asset to determine risk regimes, and a defensive Treasury allocation during unfavorable regimes.

Main conclusion. The best full-sample extension is **Sector Top-2 SHY Net**: it holds the two strongest offensive sectors when the regime is risk-on and switches entirely to SHY when the regime is risk-off. It delivers the highest full-sample CAGR among the tested strategies and materially lower drawdown than buy-and-hold equity benchmarks. However, the evidence is deliberately interpreted conservatively: walk-forward model selection does not always recover the full-sample winner, and bootstrap intervals do not establish statistically decisive raw outperformance over SPY or VTI. The strongest finding is risk-based: the defensive risk-off rule improves the volatility profile of sector momentum strategies while keeping competitive returns.

The project is best described as a **replication plus extension**: it is not claiming to invent HAA or momentum investing. It replicates the core logic of the reference market-cap strategy, then investigates whether the same framework remains useful in a more granular sector universe.

2 Relation to the Reference Paper

The reference paper is Peter B. Richman’s *A Regime-Aware Market Capitalization Rotation Strategy Using Hybrid Asset Allocation Momentum* (2026). The paper applies a HAA-style momentum framework to U.S. market-capitalization rotation. Its universe is:

- SPY, MDY, IJR as offensive assets representing large-, mid-, and small-cap U.S. equities;
- TIP as the canary asset;
- SHY as the defensive allocation;
- VTI as the benchmark.

The reference paper uses a multi-horizon momentum score called 13612:

$$M_t = \frac{1}{4} (R_{1m,t} + R_{3m,t} + R_{6m,t} + R_{12m,t}),$$

where $R_{h,t}$ is the trailing total return over h months. If the 13612 momentum of TIP is non-positive, the portfolio shifts to SHY. If it is positive, the portfolio allocates to the strongest market-cap ETF by the same 13612 momentum score.

What this repository adds

The repository extends the original paper in four directions:

1. It reproduces the HAA-style market-cap rotation logic in Python.
2. It extends the framework from market-cap rotation to U.S. sector ETF rotation.
3. It tests several sector variants: Top-1, Top-2, and Top-2 with SHY as risk-off asset.
4. It adds robustness diagnostics that are not central in the reference paper: walk-forward model selection, paired block bootstrap inference, canary sensitivity, and ETF proxy factor exposure analysis.

This positioning matters. The contribution is not “a new HAA framework”. The contribution is an empirical extension and a more conservative robustness analysis.

3 Model Architecture

3.1 Regime signal

The model computes the 13612 momentum score for the canary asset, usually TIP. The regime is defined as:

$$\text{Risk-on at } t \iff M_t(\text{canary}) > 0.$$

If the canary score is non-positive, the strategy enters a defensive risk-off regime.

3.2 Implementation timing

Signals are computed at the end of month t and implemented for month $t + 1$. This convention is essential because it prevents a direct look-ahead bias: the portfolio return for month $t + 1$ is generated using only information available at month-end t .

3.3 Transaction costs

The strategy applies a simple one-way turnover cost. Turnover is defined as:

$$\text{turnover}_t = \frac{1}{2} \sum_i |w_{t,i} - w_{t-1,i}|.$$

Net returns are computed as:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \times \text{turnover}_t,$$

where $c = 5$ basis points in the baseline specification.

4 Strategies Tested

Strategy	Risk-on rule	Risk-off rule
HAA Market-Cap Net	100% in the strongest of SPY, MDY, IJR	100% in SHY
Sector Top-1 Net	100% in the strongest offensive sector	100% in the strongest defensive asset among defensive sectors and SHY
Sector Top-2 Net	50% in each of the two strongest offensive sectors	50% in each of the two strongest defensive assets
Sector Top-2 SHY Net	50% in each of the two strongest offensive sectors	100% in SHY

The main sector extension divides the sector universe into offensive and defensive assets. Offensive sectors are XLK, XLY, XLI, XLF, and XLE. Defensive sectors are XLP, XLU, and XLV. SHY is used either as a defensive candidate or as the full risk-off allocation.

5 Full-Sample Results

The full-sample backtest covers January 2005 to December 2025 at monthly frequency. The baseline transaction cost is 5 basis points per unit of one-way turnover.

Strategy	CAGR	Vol.	Sharpe	Max DD	Avg. Turnover
HAA Market-Cap Net	11.04%	13.06%	0.87	-25.83%	35.32%
Sector Top-1 Net	11.22%	18.06%	0.68	-34.25%	44.05%
Sector Top-2 Net	11.93%	15.97%	0.79	-40.67%	32.94%
Sector Top-2 SHY Net	12.09%	14.79%	0.85	-31.56%	28.77%
VTI Buy and Hold	10.60%	15.31%	0.74	-50.84%	—
SPY Buy and Hold	10.61%	14.81%	0.76	-50.78%	—

Interpretation

Sector Top-2 SHY Net has the highest full-sample CAGR. It also has a much smaller maximum drawdown than **SPY** and **VTI**. However, the HAA market-cap replication remains the cleanest defensive benchmark because it has the strongest Sharpe ratio and lowest drawdown among the dynamic strategies.

The first-order interpretation is therefore not that the sector extension dominates everything. The correct interpretation is:

Sector Top-2 SHY Net increases return relative to the HAA market-cap baseline, but it is less defensive than HAA. It improves the risk profile of sector momentum variants, especially relative to more equity-exposed sector rotation rules.

6 Walk-Forward Model Selection

A major risk in empirical strategy research is selecting the best strategy after observing the full sample. To address this, the repository implements a rolling walk-forward selection protocol:

1. Use the previous 10 calendar years as the training window.
2. Select the best candidate strategy using in-sample Sharpe or CAGR.
3. Hold the selected strategy during the following test year.
4. Repeat the process from 2015 to 2025.

Strategy	Total Return	CAGR
Walk-Forward Selected by Sharpe	268.79%	12.60%
Walk-Forward Selected by CAGR	207.04%	10.74%
HAA Market-Cap Net	230.23%	11.47%
Sector Top-1 Net	134.49%	8.06%
Sector Top-2 Net	273.25%	12.72%
Sector Top-2 SHY Net	342.16%	14.47%
VTI Buy and Hold	280.30%	12.91%
SPY Buy and Hold	299.53%	13.42%

The fixed **Sector Top-2 SHY Net** rule remains the strongest over 2015–2025. But it is not immediately recovered by an honest ex-ante model selection rule. Sharpe-based selection chooses **Sector Top-2 SHY Net** only from 2022 onward. CAGR-based selection adopts it even later.

Walk-forward conclusion. The full-sample winner is attractive as a fixed allocation rule, but the timing of model selection is unstable. This limits any strong claim that the model would have been obviously selected in real time.

7 Bootstrap Significance Analysis

The repository uses a paired circular block bootstrap with 10,000 samples and a 6-month block length. The goal is to test whether the observed performance differences are robust to sampling variation.

Benchmark	CAGR diff.	Sharpe diff.	Interpretation
SPY Buy and Hold	+1.47%	+0.090	Confidence interval includes zero
VTI Buy and Hold	+1.48%	+0.111	Confidence interval includes zero
HAA Market-Cap Net	+1.05%	-0.021	Confidence interval includes zero
Sector Top-2 Net	+0.16%	+0.060	Confidence interval includes zero
Sector Top-1 Net	+0.87%	+0.168	Confidence interval includes zero

Interpretation

The raw performance advantage of **Sector Top-2 SHY Net** over **SPY** and **VTI** is positive, but it is not statistically decisive under the bootstrap intervals. This does not invalidate the strategy. It means that the project should not claim a statistically proven return anomaly.

The strongest bootstrap result is instead risk-based:

- **Sector Top-2 SHY Net** has significantly lower volatility than **Sector Top-2 Net**.
- **Sector Top-2 SHY Net** has significantly lower volatility than **Sector Top-1 Net**.
- **Sector Top-2 SHY Net** is significantly more volatile than **HAA Market-Cap Net**.

8 ETF Proxy Factor Exposure Analysis

The factor analysis tests whether the strategy is merely a disguised equity beta exposure. The model uses ETF-based proxy factors rather than formal academic Fama-French or Carhart factors. This is why the analysis should be interpreted as diagnostic, not as a definitive asset-pricing test.

Proxy factors used

- **MKT_SPY**: broad equity market exposure;
- **SHY**: short-duration Treasury exposure;
- **SIZE_IJR_MINUS_SPY**: small-cap proxy;
- **MID_MDY_MINUS_SPY**: mid-cap proxy;
- **TIP_MINUS_SHY**: inflation-linked bond proxy;
- **OFFENSIVE_SECTORS_MINUS_SPY**: offensive sector basket proxy;
- **DEFENSIVE_SECTORS_MINUS_SPY**: defensive sector basket proxy.

Model	Ann. alpha	Alpha t-stat	R^2	Mkt beta
Market only	4.76%	2.00	0.483	0.694
Market + defensive	8.48%	3.40	0.514	0.674
Style proxy	5.20%	2.30	0.541	0.763
Sector proxy	5.58%	2.52	0.565	0.685

Interpretation

The strategy keeps a positive intercept across ETF proxy specifications. Its market beta is below one, generally between 0.67 and 0.76. This suggests that the strategy is not simply a high-beta version of SPY.

However, these are proxy regressions. A formal asset-pricing test would require external factor datasets such as Fama-French and Carhart factors. The proper conclusion is therefore:

The ETF proxy regressions suggest that the allocation mechanism contributes to returns beyond passive equity exposure. They do not prove a persistent academic alpha.

9 Canary Robustness

The model is sensitive to the choice of canary asset.

Canary	CAGR	Vol.	Sharpe	Max DD	Risk-on freq.
TIP	12.09%	14.79%	0.85	-31.56%	75.40%
SPY	11.69%	12.67%	0.94	-20.25%	83.33%
VTI	10.72%	12.50%	0.88	-20.25%	81.75%
SHY	10.58%	16.46%	0.70	-54.21%	85.32%

TIP produces the highest CAGR and remains aligned with the original HAA logic. SPY gives the best Sharpe and drawdown profile. SHY performs poorly as a canary, even though it is useful as a defensive asset. This shows that a defensive asset is not necessarily a good regime proxy.

10 Overall Conclusions

The project has three main conclusions.

1. The extension is economically interesting

Moving from market-cap rotation to sector rotation creates a stronger full-sample return profile. The strongest variant is **Sector Top-2 SHY Net**, which preserves sector momentum exposure in risk-on regimes and uses SHY as a clean defensive allocation in risk-off regimes.

2. The evidence must be interpreted conservatively

The strategy's raw full-sample outperformance over SPY and VTI is not statistically decisive under bootstrap confidence intervals. Walk-forward selection also shows that the best full-sample rule is not automatically selected in real time.

3. The strongest contribution is risk-profile improvement

The clearest empirical contribution is not a proven return anomaly. It is the improvement of the risk profile of sector momentum strategies. **Sector Top-2 SHY Net** reduces volatility relative to more equity-exposed sector variants while keeping competitive returns and positive ETF proxy alpha estimates.

11 Limitations and Next Steps

Current limitations

- Data come from `yfinance`, not an institutional total-return database.

- The universe is limited to U.S. ETFs.
- Transaction costs are modeled simply as a flat 5 bps turnover cost.
- Bootstrap inference depends on block length assumptions.
- The factor model uses ETF proxy factors, not formal Fama-French or Carhart factors.
- The strategy has not yet been corrected for multiple testing with methods such as White Reality Check or Deflated Sharpe Ratio.

Recommended next steps

1. Add formal Fama-French/Carhart factor regressions.
2. Add stress-period tables for 2008, 2020, and 2022.
3. Add unit tests and GitHub Actions continuous integration.
4. Add a Deflated Sharpe Ratio or White Reality Check to address multiple-testing risk.
5. Test the method on additional ETF universes or non-U.S. markets.

12 How to Present the Project

A concise CV description would be:

Regime-Aware Sector Rotation Research Project. Reproduced and extended a HAA market-capitalization rotation paper in Python by applying the same momentum and canary-asset framework to U.S. sector ETFs. Implemented transaction-cost-adjusted backtests, walk-forward model selection, paired block bootstrap inference, canary robustness tests, and ETF proxy factor regressions. Found that defensive sector rotation improves the risk profile of sector momentum strategies, while raw outperformance over broad equity benchmarks remains statistically nuanced.

References

- Richman, P. B. (2026). *A Regime-Aware Market Capitalization Rotation Strategy Using Hybrid Asset Allocation Momentum*. SSRN working paper.
- Faber, M. T. (2007). A Quantitative Approach to Tactical Asset Allocation. *The Journal of Wealth Management*.
- Fama, E. F., & French, K. R. (1992). The Cross-Section of Expected Stock Returns. *The Journal of Finance*.
- Jegadeesh, N., & Titman, S. (1993). Returns to Buying Winners and Selling Losers. *The Journal of Finance*.
- Keller, W., & Keuning, J. W. (2016, 2023). Hybrid Asset Allocation and canary universe papers.